

Authored by : Arpit Raj , Lnmiit jaipur

Database Constraints Notes

Drop Constraint

Drop PRIMARY KEY

```
ALTER TABLE Students  
DROP PRIMARY KEY;
```

Cannot drop it while dependent foreign keys exist.

Drop FOREIGN KEY (MySQL)

```
ALTER TABLE Orders  
DROP FOREIGN KEY fk_customer;
```

Drop CHECK

```
ALTER TABLE Employees  
DROP CHECK chk_salary;
```

Drop UNIQUE

In MySQL, a UNIQUE constraint is backed by a unique index.

```
ALTER TABLE Users  
DROP INDEX uq_email;
```

Rename Constraint

MySQL

MySQL does not provide a general RENAME CONSTRAINT statement.
Usually you:

- Drop the old constraint.

- Create it again with the new name.

SQL Server

Supports renaming via stored procedures.

PostgreSQL

```
ALTER TABLE Employees  
RENAME CONSTRAINT  
old_name TO new_name;
```

11.4 Enable / Disable Constraints

Useful during:

- Bulk imports
- Data migration

MySQL

Foreign keys

```
SET FOREIGN_KEY_CHECKS=0;
```

Later

```
SET FOREIGN_KEY_CHECKS=1;
```

SQL Server

```
ALTER TABLE Employees  
NOCHECK CONSTRAINT ALL;
```

Enable

```
CHECK CONSTRAINT ALL;
```

Migration Strategies

Typical production workflow



Why do named constraints matter?

They make it much easier to:

- drop constraints,
- modify them,
- manage database migrations.

How Constraints are Stored

Constraints are metadata.
They are not stored inside every row.
Instead, database keeps them in internal catalogs.

Think of it like

Table Data + Metadata

Metadata describes:

- columns
- indexes
- constraints
- defaults
- foreign keys

MySQL Information Schema

Examples

```
SHOW CREATE TABLE Students;
```

shows the table definition, including constraints.

Constraints and Indexes

Constraint	Index Creation / Rules
PRIMARY KEY	↓ Automatically creates Unique Index.
UNIQUE	

	↓ Automatically creates Unique Index.
FOREIGN KEY	Needs Referenced column ↓ Indexed. In MySQL InnoDB: • The parent key must be indexed (typically PRIMARY KEY or UNIQUE). • An index on the child foreign key is automatically created if one does not already exist.
CHECK	No index. Simply evaluates expression.
DEFAULT	No index.
NOT NULL	No index.

Why do constraints help query optimization?

Because they provide the optimizer with guarantees about the data, allowing it to generate more efficient execution plans.